Minseok Kwak

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EDUCATION

University College London London, UK
MSc Artificial Intelligence & Medical Imaging Sept 2026 – Sept 2027
Department of Medical Physics & Biomedical Engineering
Core Modules: Inverse Problems in Imaging, Medical Imaging with Ionising Radiation, Biomedical Ultrasound, Information Processing in Medical Imaging, Computer-Assisted Surgery and Therapy, Programming Foundations for Medical Image Analysis, Machine Learning in Medical Imaging, Applied AI in Medical Imaging
BSc Computer Science (Graduate of 2026) Sept 2020 – Sept 2026

EXPERIENCE

Seoul National University – Visual & Geometric Intelligence Lab Seoul, South Korea
Machine Learning Intern June 2025 – Jan 2026
- Developed deep learning models and applied enhanced implementation methods to system architecture
- Ran state-of-the-art LLM models and optimised training datasets through hyperparameter tuning
- Submitted a paper to CVPR/ECCV 2026 as third author: new dataset and pipeline for VLMs achieving an average accuracy improvement exceeding 40% over the best-performing open-source models

National Autistic Society & MotionInputGames London, UK
Lead Developer Sept 2024 – May 2025
- Developed a motion-based game for autistic children as a therapeutic tool, fostering self-oriented motor skills
- Utilised Unity 3D and Stable Diffusion as primary development tools
- Optimised in-game logic, physics, and rendering/GPU algorithms

Capital Mechanised Infantry Tiger Division, ROK Army Gapyeong, South Korea
Network Communications Specialist Nov 2022 – May 2024
- Completed 18 months of military service specialising in network security and long-range communication networks
- Maintained secure data-handling procedures for sensitive military classifications

PROJECTS

SpatialMosaic: A Multiview VLM Dataset for Partial Visibility
CVPR/ECCV 2026 Submission (Third Author) Jul 2025 – Feb 2026
- Developed a comprehensive instruction-tuning dataset featuring 2M QA pairs targeting realistic multi-view scenarios with partial visibility, object occlusion, and sparse visual cues – a gap absent from existing VLM benchmarks
- Integrated 3D foundation model VGGT to introduce SpatialMosaicVLM, a hybrid framework combining visual encoding with spatial reasoning; available on arXiv
- Ranked 1st on both SpatialMosaic-Bench and VSTI-Bench, exceeding 40% average accuracy improvement over the best-performing open-source models and demonstrating strong zero-shot generalisation to spatial-temporal tasks

SEAMLESS: Specularity-Aware End-to-End Adaptive Masking and Luminance Estimation via Self-Supervision
IPCAI 2027 Submission (First Author) Nov 2025 – Oct 2026
- Proposed a fully self-supervised, end-to-end illumination decomposition for endoscopic depth and pose estimation that separates luminance, specularities, and tissue albedo, removing the reliance on handcrafted specularity detectors in the prior state of the art (SHADeS)
- To our knowledge, the first monocular depth model in endoscopy that explicitly learns to detect and separate specular reflections through light decomposition
- Outperformed SHADeS on every depth metric on C3VD (AbsRel 0.1312 to 0.1104, a 15.9% reduction), with a wider margin on the more realistic C3VDv2 (0.1423 to 0.1156, 18.8%), and produced visibly cleaner depth on real EndoMapper colonoscopies

NAS Consortium – Winter Sports Children's Game
Industrial Partnership Sept 2024 – May 2025
- Developed a motion-input therapeutic game for autistic children, currently available on the Microsoft Store
- Facilitated weekly cross-stakeholder meetings with NAS therapists and client groups to align requirements across therapeutic, technical, and user-experience goals

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C#, Haskell, JavaScript
Deep Learning & Vision: PyTorch, TorchVision, CUDA, OpenCV, Open3D, COLMAP, VGGT, CLIP
Scientific Computing & Tools: NumPy, SciPy, Matplotlib, TensorBoard, Weights & Biases, Git, Conda, Docker